

The Measurement Postulate

Measurement is essential to get the result of a quantum computation. In classical computation it simply consists in reading the output register. However, it is not so straightforward in quantum computation.

In terms of physical systems, measuring means make the system interact with a measuring apparatus prepared by an observer interested in some given property of the system.

Here, the notion of **OBSERVABLE** is crucial.

Measurement Postulate

Every **observable** M (i.e. every property of a physical system, which can be measured) is represented by a Hermitian operator on the system states, and viceversa.

To understand the representation and the meaning of an observable we first recall the notion of **external product** and the representation of a linear operators by means of it.

$$|v\rangle \in V, |w\rangle \in W$$

$$|w\rangle\langle v| : V \rightarrow W$$

is the linear operator returning for all $v' \in V$, $\langle v|v'\rangle|w\rangle$

Given an orthonormal basis for V , $\{|i\rangle\}_i$, any $v \in V$ can be expressed as $\sum_i v_i |i\rangle = \left(\sum_i |i\rangle\langle i|\right)v \Rightarrow \sum_i |i\rangle\langle i| = I$

The operators $P_i = |i\rangle\langle i|$ are called orthogonal **Projectors** and for any decomposition of the identity $I = \sum_i P_i$ we can define a **projective measurement** with results **i** with probability $p(i) = \langle \psi | P_i | \psi \rangle$ and post-measurement state: $\frac{P_i | \psi \rangle}{\sqrt{p(i)}}$ (this is also called von Neumann measurement)

Example: $P_0 = |0\rangle\langle 0|$ and $P_1 = |1\rangle\langle 1|$ are the projectors associated to the orthogonal basis $\{|0\rangle, |1\rangle\}$ of 1 qubit. Measuring $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ in the computational basis means applying a projective measurement with results

0	with prob.	$\langle \psi 0 \rangle \langle 0 \psi \rangle$	$= \alpha^* \alpha = \alpha ^2$
1	with prob.	$\langle \psi 1 \rangle \langle 1 \psi \rangle$	$= \beta^* \beta = \beta ^2$

In general, any Hermitian operator (observable) M has a **spectral decomposition**

$$M = \sum_i m_i P_i$$

where P_i are the projectors on the eigenspaces of the eigenvalues m_i of M . Therefore, measuring the observable M means performing a projective measurement with respect to the orthonormal basis of the eigenvalues of M .

Example $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ is Hermitian with spectral decomposition $Z = |0\rangle\langle 0| - |1\rangle\langle 1|$

In fact, $+1$ and -1 are its eigenvalues wrt the eigenvectors $|0\rangle$ and $|1\rangle$: $Z|0\rangle = +1|0\rangle$ and $Z|1\rangle = (-1)|1\rangle$

X and Y are also Hermitian operators with eigenvalues ± 1 with respect to $\{|+\rangle, |-\rangle\}$ and $\left\{\frac{|0\rangle + i|1\rangle}{\sqrt{2}}, \frac{|0\rangle - i|1\rangle}{\sqrt{2}}\right\}$, respectively.

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad Y \cdot \begin{pmatrix} \frac{1}{\sqrt{2}} \\ i\frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ i\frac{1}{\sqrt{2}} \end{pmatrix} \quad Y \cdot \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -i\frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} -\frac{1}{\sqrt{2}} \\ i\frac{1}{\sqrt{2}} \end{pmatrix} = (-1) \begin{pmatrix} \frac{1}{\sqrt{2}} \\ i\frac{1}{\sqrt{2}} \end{pmatrix}$$

Thus, a diagonal representation of Y is wrt basis $\left\{\frac{|0\rangle + i|1\rangle}{\sqrt{2}}, \frac{|0\rangle - i|1\rangle}{\sqrt{2}}\right\}$

SPECTRAL THEOREM

Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Then the following are equivalent:

- (a) T is normal.
- (b) V has an orthonormal basis consisting of eigenvectors of T .
- (c) T has a diagonal matrix with respect to some orthonormal basis of V .